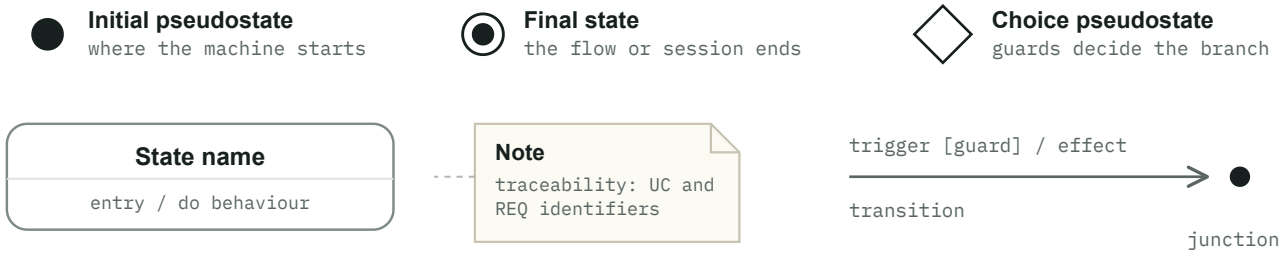
PathSG Dialog Map

Every screen, sub-state, dialog and transition in the PathSG interface, drawn as UML state machine diagrams — one per feature area. Each state is something the user sees; each transition is labelled `trigger [guard] / effect`, and traceability to the use cases and requirements is carried in attached UML notes.

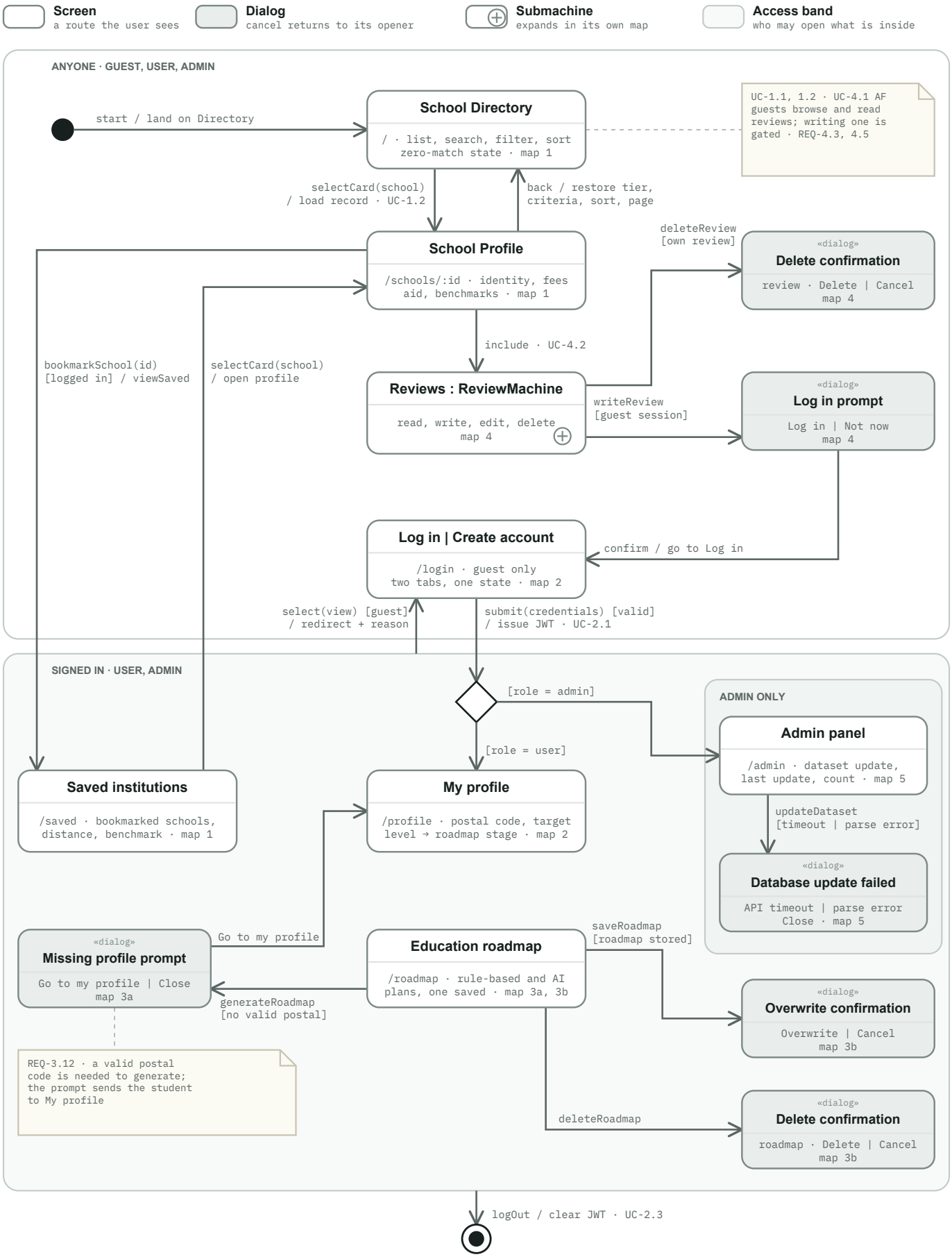
UC IDs refer to the Use Case Descriptions; **REQ** IDs to the functional requirements; PERF / SEC / QUAL / BR IDs to the non-functional requirements and business rules.



A composite state split by a dashed divider holds orthogonal (concurrent) regions — see map 3a.

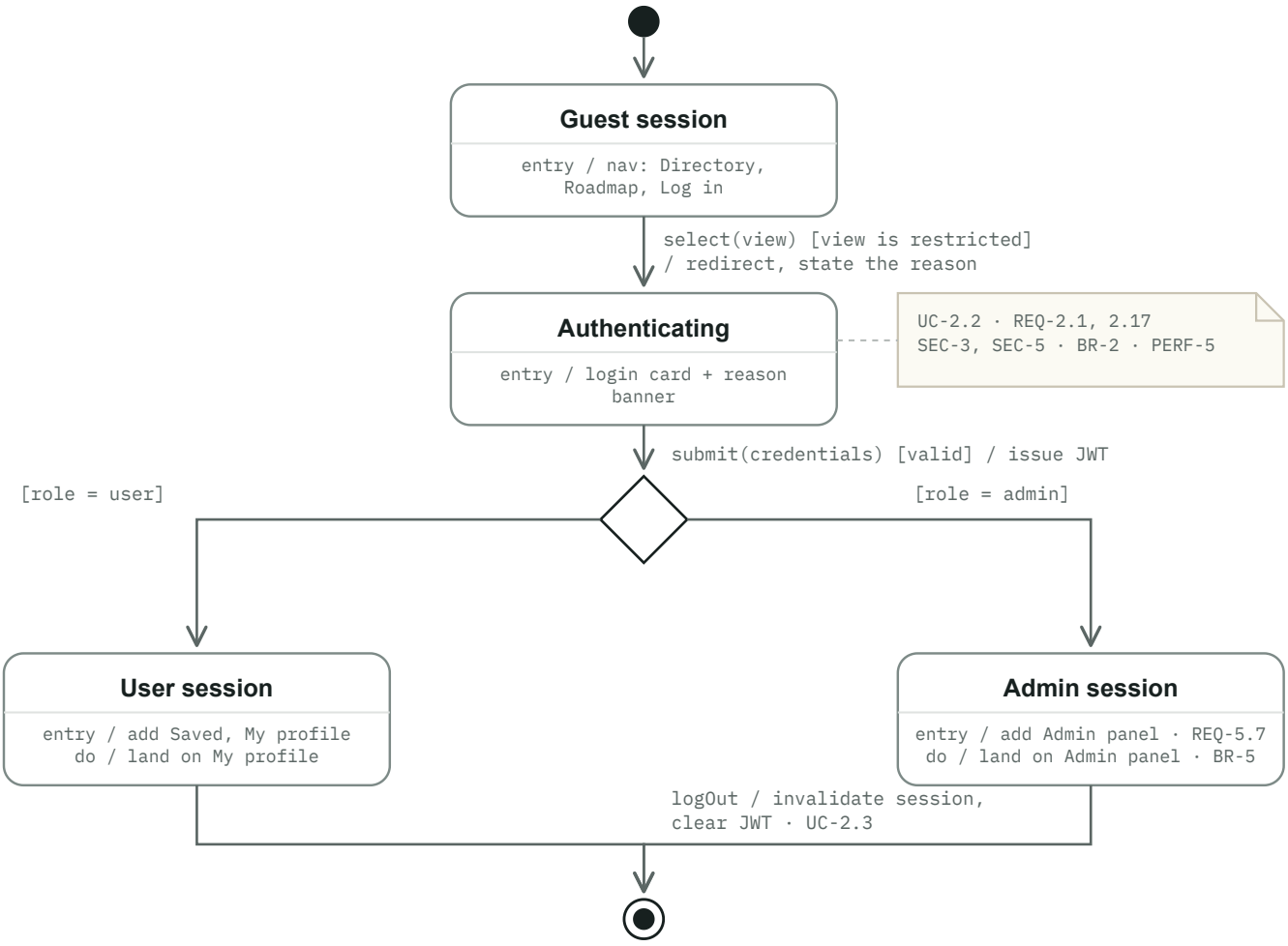
M · Master dialog map

Every screen and dialog on one page, joined by the transitions that cross between maps 0 to 5. Bands group screens by who may open them. Header links, which already reach every screen a role can see, are left out. Each state names the map that expands it and links to it.



0 · Shell and access control

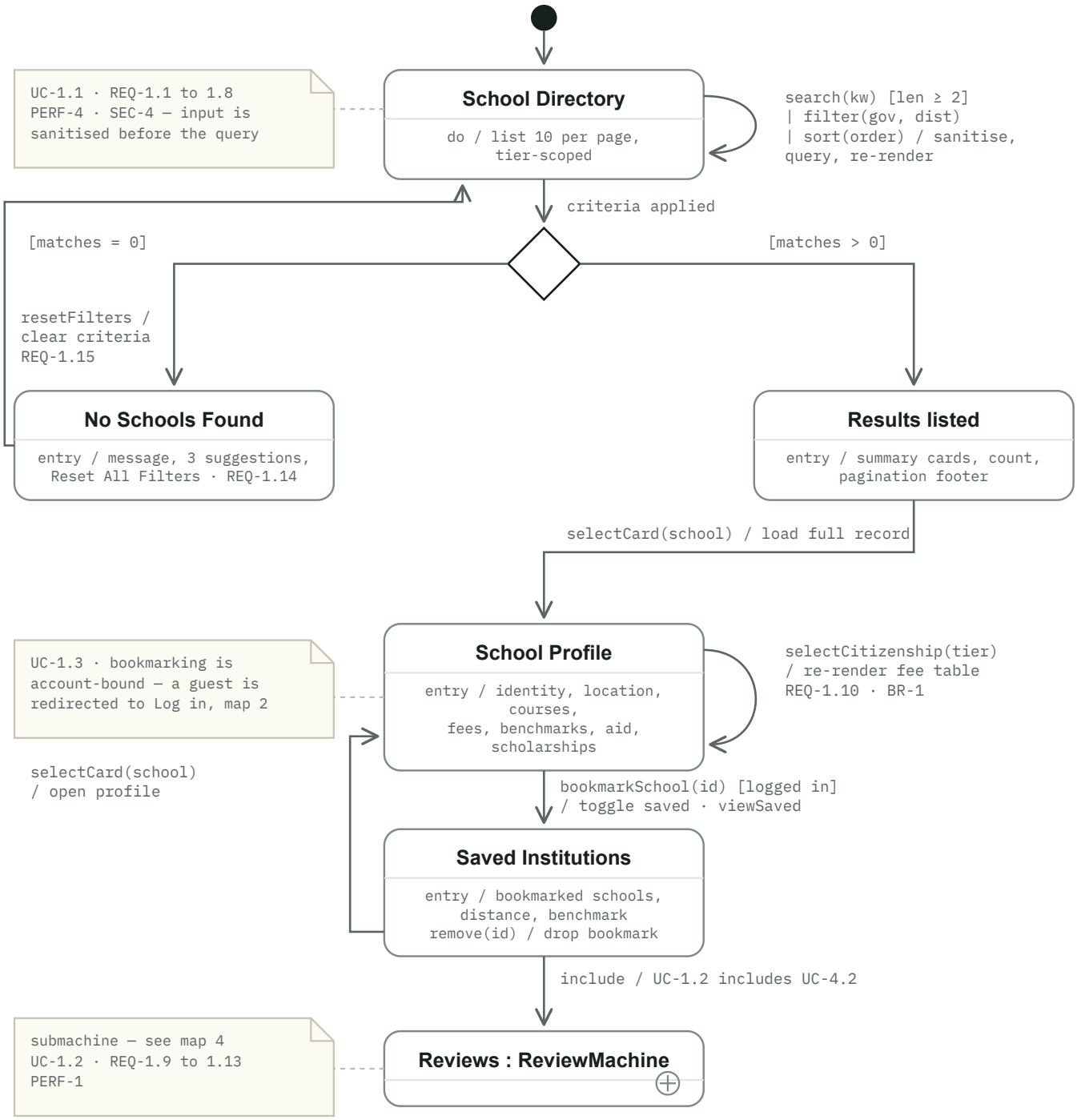
One persistent header carries every map that follows, and the role held by the session decides both what it shows and where login lands. Directory and Roadmap are always present; the authenticated items appear on login, and the Admin panel item only for the Admin role.



SCREEN OR ACTION	GUEST	USER	ADMIN	ENFORCED BY
Directory, School profile	Full	Full	Full	–
Reviews — read	Full	Full	Full	REQ-4.3
Reviews — write, edit, delete	Gated by dialog	Own review only	Own review only	REQ-4.5, 4.10 · BR-6
Bookmark control on cards	Hidden	Full	Full	BR-2 · REQ-2.11
Roadmap, Saved, My profile	Redirect to login	Full	Full	REQ-2.17 · BR-2 · SEC-5
Admin panel	Hidden	Hidden	Full	REQ-5.1, 5.7 · BR-5

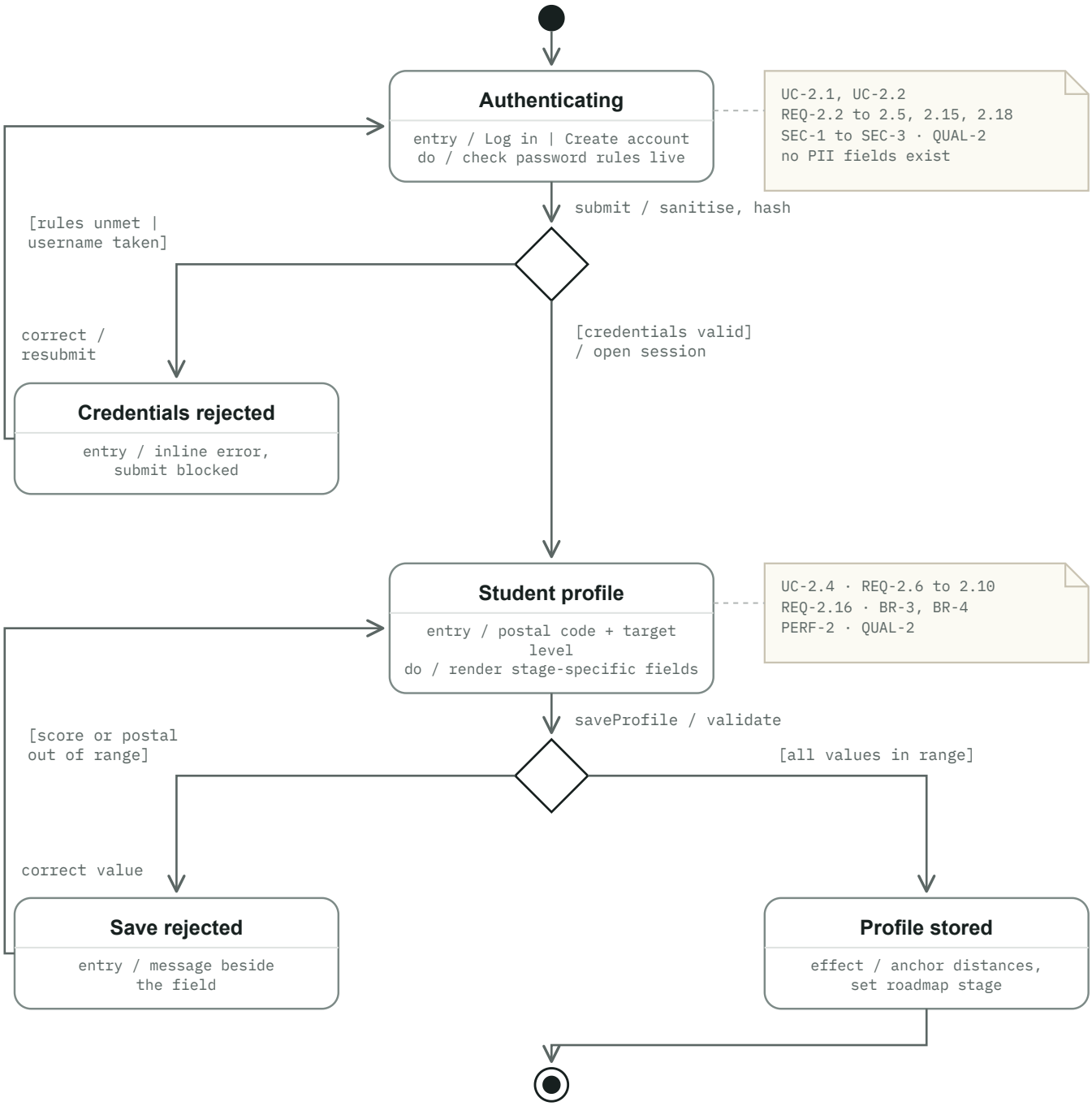
1 · School Directory and profile

Search, filter and sort are internal transitions on one state, so the alternative flows — zero matches and malformed input — resolve through a choice rather than separate pages. Selecting a card is the only transition that leaves the directory; `back` restores the tier, criteria, sort order and page. Bookmarking sits after the profile: the toggle is account-bound, so a guest is sent to Log in, and the saved set is its own state a card can be reopened from.



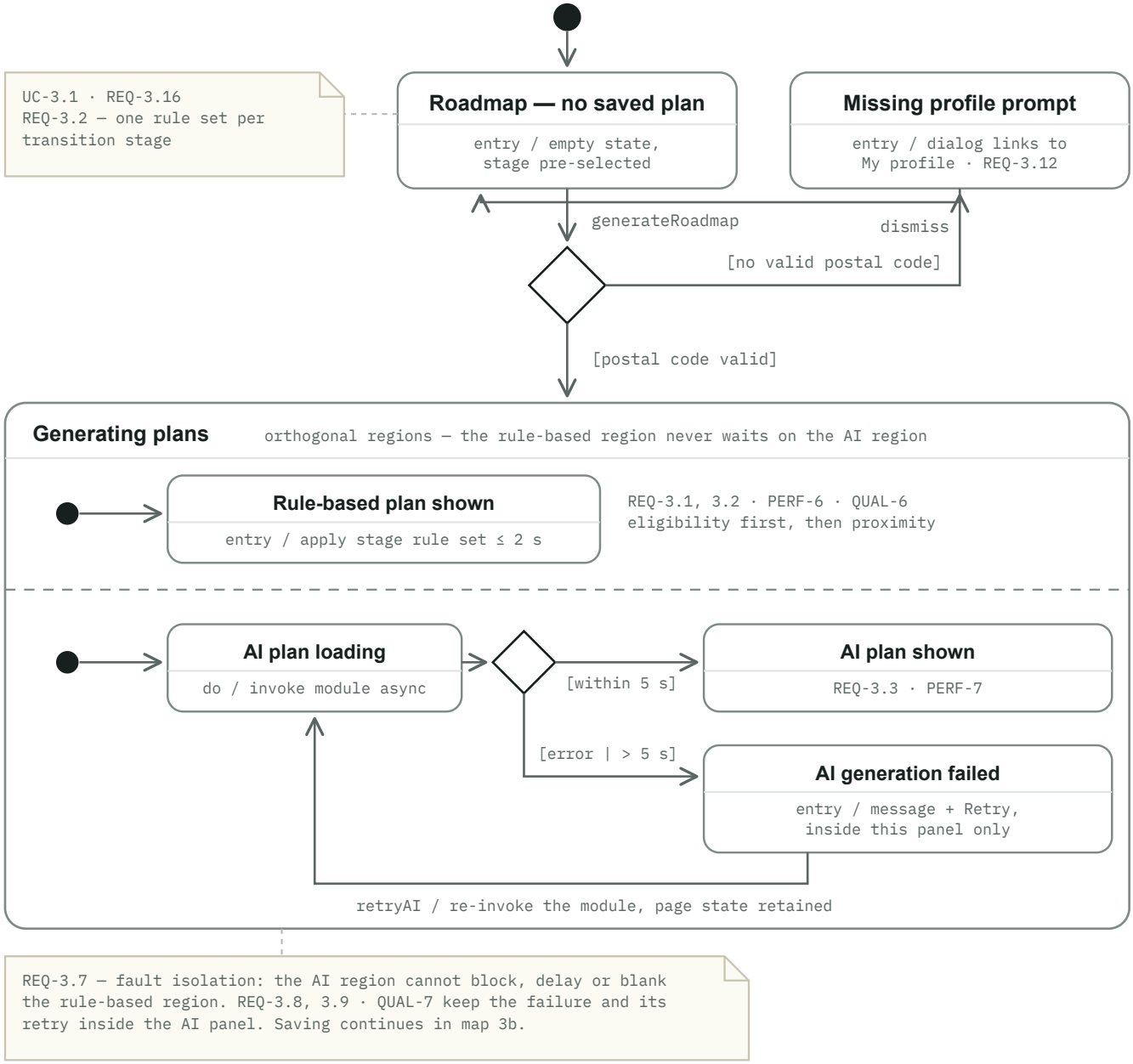
2 · Account and student profile

Log in and Create account are tabs of one state, so both paths reach the same profile. That profile is the data spine of the system: its postal code anchors every distance calculation and its target level pre-selects the roadmap stage, and neither can be stored out of range.



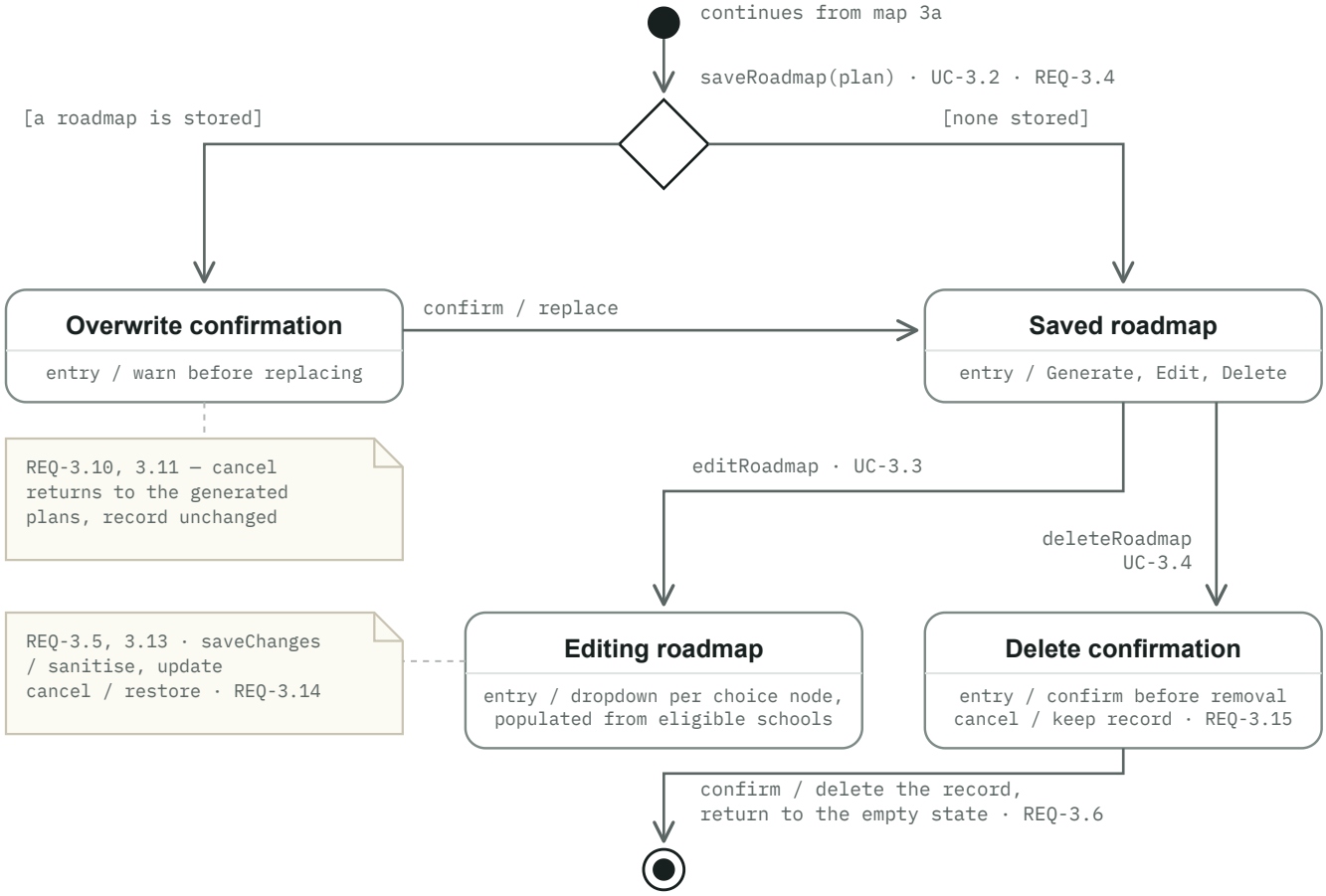
3a · Roadmap generation

Generation is modelled as a composite state with two orthogonal regions, because that is exactly the guarantee the requirements ask for: the rule-based region completes on its own clock, and any fault in the AI region stays inside that region with a retry of its own.



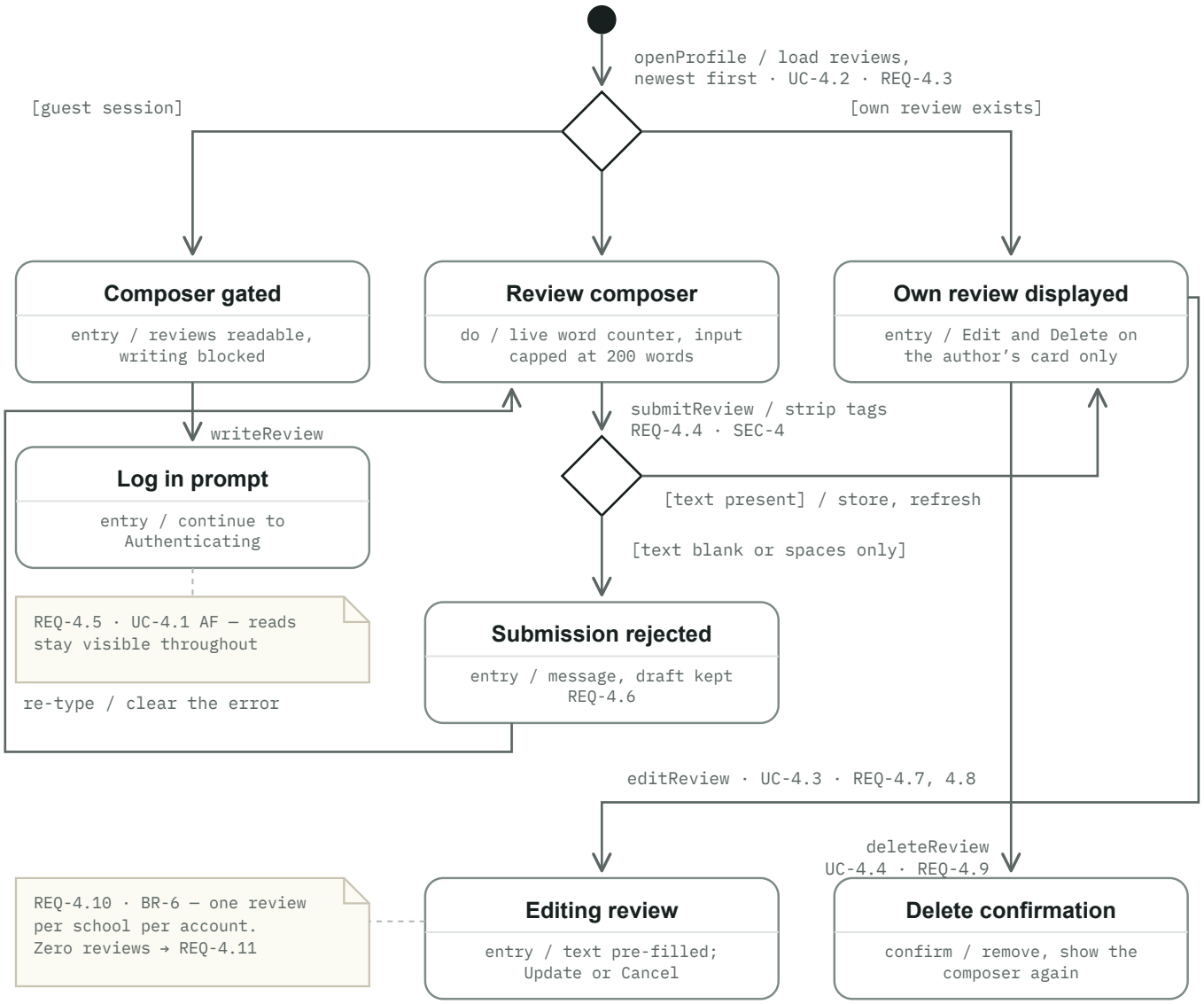
3b · Saving, editing and deleting a roadmap

One roadmap may be stored per account, which makes the overwrite confirmation part of the ordinary save path rather than an edge case. Cancelling anywhere leaves the stored record untouched.



4 · School reviews

The submachine referenced by the school profile. Its initial choice reads the session and the account's review history, so the same section opens gated, as a composer, or as the author's own card; editing and deleting are reachable only from that card.



5 · Admin manual database update

One state, one trigger, three terminal outcomes. The control locks for the duration of the run and both failure branches roll the transaction back, so the junction always returns the panel to a truthful idle state showing the last successful update.

